

Topic 7: Inheritance - Single and Multilevel

7.1 What is Inheritance?

- Inheritance allows a new class (derived/child) to acquire properties and behaviour from an existing class (base/parent)
- Promotes code reuse -- write common logic once in the base class
- Models an IS-A relationship: a Dog IS-A Animal, a SavingsAccount IS-A BankAccount
- Syntax: class Derived : access-specifier Base { ... };
- The most common access specifier is public -- derived class inherits public and protected members

7.2 Single Inheritance

- One derived class inheriting from one base class
- Derived class can add new members and override existing ones
- Base class members marked protected are accessible in the derived class but not outside
- Base class constructor is called before derived class constructor

```
class Animal {
protected:
    string name;
public:
    Animal(string n) : name(n) {}
    void eat() { cout << name << " is eating" << endl; }
};

class Dog : public Animal {
    string breed;
public:
    Dog(string n, string b) : Animal(n), breed(b) {}
    void bark() { cout << name << " barks!" << endl; }
};
```

7.3 Calling Base Class Constructors

- Use the initialiser list to forward arguments to the base constructor
- If no base constructor is explicitly called, the default base constructor is invoked
- Destructor order is the reverse of constructor order: derived destructor runs first, then base

```
Dog d("Rex", "Labrador");  
d.eat();    // inherited from Animal  
d.bark();   // Dog's own method
```

7.4 Multilevel Inheritance

- A chain of inheritance: A -> B -> C -- B derives from A, C derives from B
- C inherits all public/protected members from both A and B
- Constructor chain: A() runs first, then B(), then C()

```
class Shape { /* base */ };  
class Polygon : public Shape { /* adds sides */ };  
class Triangle : public Polygon { /* 3 sides */ };
```

7.5 Access Specifiers in Inheritance

- public inheritance: public -> public, protected -> protected (most common)
 - protected inheritance: public -> protected, protected -> protected
 - private inheritance: public -> private, protected -> private (rarely used)
 - Private base members are NEVER inherited regardless of access specifier
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